

Liver Biopsy in Children.

In the last two chapters we learned to read a liver biopsy. We looked at the patterns of injury, at fibrosis and at the special stains. In this chapter we step back to the bedside. We ask when a child really needs a biopsy, how to take it safely, and how to make sure that the piece of liver that reaches the pathologist can answer our question.

This matters because a liver biopsy is the most invasive test we do in hepatology. It needs sedation or anaesthesia in most children. It carries a small but real risk of bleeding, and a few children have died after it. At the same time, a biopsy that is too small, or taken at the wrong time, or sent to the laboratory in the wrong way, can give a false answer. So the risk is only worth taking if the biopsy is done for a clear reason, by the right route, with enough tissue, handled correctly.

So this chapter asks four questions. First, when is a biopsy still needed? Second, which route should we use? Third, how much tissue is enough? And fourth, who bleeds, and what should we do about blood tests before the needle?

We will start with the indications, first when the diagnosis is still open, and then when the diagnosis is already known. Then we will go through the six routes into a child's liver. Then we will look at the size of the sample and at sampling error. After that we will look at complications, at who is at risk, and at how long to watch the child. Then we will come to the most controversial part, which is what to do with the INR and the platelet count. We will finish with a practical checklist, with what the guidelines say, and with four clinical cases.

A note on the sources. The main paediatric document is the position paper of the European paediatric society, ESPGHAN, written by Dezsófi and colleagues in 2015. For adults, we will use the position paper of the American Association for the Study of Liver Diseases, or AASLD, from 2009, and its guidance on coagulation in liver disease from 2020. We will also use the joint British guideline of 2020, written by Neuberger and colleagues for the British Society of Gastroenterology, the Royal College of Radiologists and the Royal College of Pathologists, and the European guideline on the prevention of bleeding in liver disease, from EASL in 2022. For fatty liver disease in children, we will use the American guidance of 2025.

Part one. When the diagnosis is still open.

For many years, the liver biopsy was the main way to find out what was wrong with a child's liver. This has changed. The central message of the ESPGHAN paper is that genes, enzyme tests and imaging now answer many of the questions

that used to need tissue. So before every biopsy, ask yourself one question. What will the tissue tell me that I cannot find out in another way?

Let us go through the main diseases.

Biliary atresia is the most important exception. Here the biopsy is still essential. An adequate core makes the correct diagnosis in about ninety-six percent of cases. Clinical findings alone are right in only about eighty to ninety percent. But there is one firm rule. The biopsy must never delay the Kasai operation. Every week of delay makes the outcome worse. And the final answer is not the biopsy. It is the cholangiogram, the picture of the bile ducts taken during surgery.

Progressive familial intrahepatic cholestasis with a normal GGT is the next example. Here a gene panel usually makes the diagnosis. But the biopsy can still add something. Special stains for the bile salt export pump, called BSEP, and for other transporters, together with electron microscopy, can support the diagnosis. More importantly, only the biopsy can tell us whether the liver is already cirrhotic. This matters, because a surgical operation to divert the bile, called biliary diversion, is only worth doing before cirrhosis has developed. So the biopsy here is selective.

Alagille syndrome is a similar story. The loss of small bile ducts, which we call ductopenia or paucity of bile ducts, can only be shown on the biopsy. But paucity is often not yet present in young infants, and it is absent in about a quarter of biopsies even after the age of one year. So a normal number of ducts does not exclude Alagille syndrome. Genetic testing for JAG one and NOTCH two, together with the clinical criteria, comes first. The biopsy is selective.

For alpha one antitrypsin deficiency, the ESPGHAN paper is very clear. A liver biopsy is not required for the diagnosis. The diagnosis is made by the phenotype, or by testing the SERPINA one gene. The biopsy can add information about prognosis, for example whether there is bridging fibrosis or a strong ductular reaction, but it is not needed to make the diagnosis.

The same is true for lysosomal acid lipase deficiency, for congenital disorders of glycosylation, and for typical congenital hepatic fibrosis. These are diagnosed by an enzyme assay, by a glycosylation screen, and by the kidney findings. The biopsy may help to stage the liver disease, but it is not needed for the diagnosis.

Finally, acute liver failure. Here the biopsy is rarely useful. The histology does not raise the chance of finding a cause, and it does not change what we do in the first days. There is one exception. If we suspect autoimmune hepatitis or Wilson disease, and the result would change treatment, a biopsy by the transjugular route may be justified.

Part two. When the diagnosis is known.

Today, the biopsy more often answers two other questions. How bad is the disease? And is the treatment working?

Autoimmune hepatitis is the clearest case. Every child with suspected autoimmune hepatitis should have a biopsy at presentation. The biopsy confirms the diagnosis, grades the inflammation, stages the fibrosis, and helps to exclude overlap with autoimmune sclerosing cholangitis. A second biopsy is recommended before treatment is withdrawn, after at least one year of complete remission, because some children still have active inflammation in the liver even when the blood tests are normal.

In Wilson disease, the most useful tissue test is not the histology. It is the amount of copper in the liver, measured in a laboratory. A liver copper above two hundred and fifty micrograms per gram of dry weight is diagnostic. The histology itself is only supportive. The biopsy is selective. It is needed when the diagnosis is uncertain, and it is debatable when the child already has two disease-causing variants in the ATP seven B gene.

In fatty liver disease in children, the American guidance of 2025 lists four reasons to biopsy. The first is diagnostic uncertainty. The second is an ALT that is at least twice the upper limit of normal. The third is concern for advanced fibrosis. And the fourth is positive autoantibodies, where autoimmune hepatitis or drug injury must be excluded. The biopsy can then confirm steatohepatitis and stage the fibrosis.

In chronic hepatitis B, the grade of inflammation and the stage of fibrosis help us decide whether to treat, especially in the immune-active phase. In the North American Hepatitis B Research Network, ninety-eight percent of biopsies were adequate, and serious adverse events occurred in one point four percent.

After liver transplantation, the biopsy is essential when the graft is not working. It shows rejection, the return of the original disease, de novo autoimmune hepatitis, and fibrosis. Protocol biopsies, taken at fixed times in children who seem well, are still debated. They are used more often in children than in adults.

Finally, a liver mass. In hepatoblastoma, the histology helps to decide the risk group, and therefore the treatment. For hepatocellular carcinoma, a biopsy is done only when imaging cannot make the diagnosis, because tumour cells can be carried along the needle track. This is called needle-track seeding, and it happens in about one point six to five percent of cases.

Part three. Six ways into a child's liver.

Once we have decided that a biopsy is needed, we must choose the route. There is a simple principle. The route is chosen by three things: the child's clotting, the anatomy, and the question we want to answer. It should not be chosen by habit.

There are six main routes.

The first is the blind percutaneous biopsy, also called the percussion-guided biopsy. The operator finds the liver by tapping on the chest, and passes the needle between the ribs in the right mid-axillary line, usually between the seventh and ninth intercostal spaces. This is only acceptable for diffuse liver disease, when an earlier ultrasound has shown normal anatomy. Taking two cores may improve the yield, but more than three passes adds minor complications. In a randomised trial of one hundred and two children in Egypt, published by Mogahed and colleagues in 2016, blind biopsy caused more serious events than biopsy with ultrasound assistance.

The second is the ultrasound-assisted or ultrasound-guided biopsy. In the assisted version, the operator marks the site with ultrasound just before the biopsy. In the guided version, the needle is watched in real time. This is the default route today. It keeps the needle away from the gallbladder, large vessels, the colon and the lung. After a segmental liver graft, such as a left lateral segment, image guidance is mandatory, because the normal landmarks do not apply. In Coimbra, Matos and colleagues reported five hundred and thirteen ultrasound-guided biopsies in children in 2012. Ninety-nine point three percent were diagnostic. Major complications occurred in one percent, and none needed surgery.

The third is the plugged percutaneous biopsy. This is a normal percutaneous biopsy, but as the needle is withdrawn, a plug of gelatin sponge, collagen or thrombin is pushed into the track to seal it. It is used when the child has a clotting problem, ascites, or is a small infant. In a series from Lungren and colleagues in 2016, sixty-seven infants weighing less than ten kilograms had a plugged biopsy. All the samples were adequate, and there was one complication.

The fourth is the transjugular biopsy. Here an interventional radiologist passes a catheter from the jugular vein in the neck, through the heart, into a hepatic vein, and takes the biopsy from inside the vein. Because the needle does not cross the liver capsule, any bleeding goes back into the circulation. This route is chosen when the INR or the platelet count is out of range, when there is ascites, or when we also want to measure the pressure in the portal system, called the hepatic venous pressure gradient. It needs general anaesthesia and an interventional

radiology suite. The complications are different: bleeding in the neck, and heart rhythm problems as the wire passes through the heart.

The fifth is the laparoscopic or open surgical biopsy. This is used when a percutaneous biopsy has failed, for a mass, or when a large piece of tissue is needed, for example for enzyme studies. But there are two problems. A wedge taken from just under the capsule overstages fibrosis, because the tissue under the capsule is normally more fibrous. And cautery can damage the tissue. In the Children's Oncology Group study of hepatoblastoma, thirty-six percent of children who had an open biopsy needed a blood transfusion, and twenty-four percent of those who had a laparoscopic biopsy.

The sixth is the biopsy guided by endoscopic ultrasound, or EUS. The needle is passed through the wall of the stomach or duodenum. This is useful when the child is already having an endoscopy, when we want tissue from both lobes, or when obesity makes the percutaneous route difficult. In Cincinnati, Schwartz and colleagues reported eighty-three of these biopsies in 2025. The median number of portal tracts was twenty-four, and ninety-three percent met the American standard for adequacy. Five percent had mild adverse events, and there was no bleeding.

Part four. The transjugular route, and how much tissue is enough.

The transjugular route deserves a closer look, because it is often chosen for the sickest children.

In 2023, Smayra and colleagues pooled eight studies with three hundred and seventy-four transjugular biopsies in children. The procedure was technically successful in ninety-nine point one percent, and the tissue was called adequate in ninety-seven point five percent. That sounds excellent. But each centre used its own definition of adequate.

A study from Lucknow in India, published by Srivastava and colleagues in 2020, gives a more honest picture. It included five hundred and forty biopsies, and twelve percent of the children needed the transjugular route. The median number of complete portal tracts was four in the transjugular cores and five in the percutaneous cores. The tissue was adequate for a full report in eighty-four percent of transjugular biopsies and ninety-seven percent of percutaneous ones. Cirrhosis and the transjugular route both predicted a core with fewer than six portal tracts. And in a large review of seven thousand six hundred and forty-nine transjugular biopsies, by Kalambokis and colleagues in 2007, complications were more frequent in children than in adults.

So the transjugular route is safer for bleeding, but the tissue is thinner. The British guideline makes a strong recommendation, based on high-quality evidence: take at least two cores by the transjugular route.

This brings us to the key question of adequacy. How much tissue is enough?

The answer is not really about length. It is about the number of complete portal tracts. The American standard asks for a core of at least two centimetres with at least eleven complete portal tracts. The Royal College of Pathologists sets a minimum of ten millimetres and six portal tracts. The ESPGHAN paper asks for a core of about twenty millimetres long and one point eight millimetres wide, and at least ten portal tracts in suspected biliary atresia.

These numbers can be joined into one simple rule. Six portal tracts can name a disease. Eleven are needed to stage it.

Now let us look at what children's cores actually contain. In the Lucknow study, the median was four portal tracts by the transjugular route and five by the percutaneous route. In a blind biopsy series from Mississippi, in children older than five years, the mean was nine point eight. With an eighteen gauge needle guided by ultrasound in Athens, the mean was eleven. In a randomised trial in liver grafts from Beijing, an eighteen gauge needle gave eighteen portal tracts and a sixteen gauge needle gave twenty. And with endoscopic ultrasound, taking cores from both lobes, the median was twenty-four.

Notice that many of these series fall below eleven. Most paediatric cores are only about one to one point one centimetres long. So many paediatric biopsies can make a diagnosis, but cannot reliably stage fibrosis.

Part five. Why the first core fails, and when the core misleads.

About one in five biopsies in children does not give a diagnosis. In a study from Los Angeles, published by Short and colleagues in 2013, nineteen point two percent of three hundred and twenty-eight first percutaneous biopsies were non-diagnostic.

Why? There are three places where a biopsy can fail. The first is the needle. The non-diagnostic cores were shorter, one point four centimetres on average, against one point seven for the diagnostic ones. The second is the indication. Biopsies taken for unexplained raised transaminases were non-diagnostic in thirty-four point nine percent, against sixteen point seven percent for other reasons. When the question is vague, the answer is often vague too. The third is the laboratory. In 2021, Liang and colleagues ran a quality improvement programme in the pathology laboratory. Before the programme, fifty-nine percent of cores were broken into fragments during processing. After it, only twenty-four percent were.

And the share of adequate cores rose from thirty-two to fifty-six percent, without any change at the bedside.

The needle matters as well. In the Beijing trial, a sixteen gauge needle gave more portal tracts than an eighteen gauge needle, and did not cause more bleeding. The ESPGHAN paper also supports the sixteen gauge needle for routine ultrasound-guided biopsy.

So before you blame the biopsy, ask three questions. How long was the core? How many portal tracts reached the slide? And why was the biopsy taken in the first place?

Even a good core can mislead. This is called sampling error, and it is predictable. It clusters in young infants, in patchy disease, and near the capsule.

In biliary atresia, the typical features may be missing if the biopsy is taken before four to six weeks of age. And other conditions can look like biliary atresia on the biopsy, such as parenteral nutrition, alpha one antitrypsin deficiency of the ZZ type, and cystic fibrosis. So if an early biopsy is equivocal, repeat it or go to the cholangiogram, and make sure the core contains at least ten portal tracts.

In Alagille syndrome, as we said, paucity may be absent in young infants and in a quarter of biopsies after the first year. A stain for cytokeratin seven, called CK seven, helps to separate the true bile ducts from the reactive ductules. But genetics should come first.

In alpha one antitrypsin deficiency, the typical globules seen with the PAS stain after diastase are often not yet visible in early infancy. So the phenotype or genotype decides, not the histology.

In acute liver failure, necrosis and regeneration are patchy. One core may show massive necrosis and another may show regenerating liver. It can also be very hard to tell acute from chronic disease. So biopsy only if autoimmune hepatitis or Wilson disease would change the treatment.

For staging fibrosis, Regev and colleagues took biopsies from both the right and the left lobe of the same adults with hepatitis C. The stage differed by at least one step in thirty-three percent. And wedge biopsies from under the capsule overstage the fibrosis. So use a deep needle core, not the capsule.

After a segmental liver graft, the landmarks are unreliable, so image guidance is needed every time.

And for a focal lesion, remember needle-track seeding. Pass the needle through a rim of normal liver on the way to the lesion, and sample both the lesion and the background liver.

Part six. Complications, and who bleeds.

How often do children bleed after a percutaneous liver biopsy? The honest answer is that each series defines a complication in its own way. But the numbers are in a narrow range.

In Coimbra, major complications occurred in one percent of five hundred and thirteen ultrasound-guided biopsies. In Oslo, in two hundred and seventy-five biopsies, including transplanted livers, the rate was one point one percent. In Houston, in two hundred and forty-nine percutaneous biopsies, it was two point four percent. In an older series of four hundred and sixty-nine biopsies, published by Cohen and colleagues in 1992, bleeding occurred in two point eight percent. In Los Angeles, three point three percent of children needed a transfusion. In Toronto, in sixty-five ultrasound-guided biopsies in infants under one year, it was four point six percent. And in children with cancer, one study found bleeding after fifteen percent of biopsies.

What about deaths? In the Cohen series, three of four hundred and sixty-nine children died, and all three had a malignancy or a blood disease. In Houston, one of two hundred and forty-nine. In Los Angeles, one of two hundred and thirteen children. In Oslo, none.

Put these numbers together and a clear message appears. The risk belongs less to the needle than to the child. The infant, the child with cancer, and the child with a failing liver carry most of the risk.

Let us look more closely at who bleeds.

Acute liver failure is the strongest risk factor. In the Oslo study, published by Westheim and colleagues in 2012, children and adults with acute liver failure had an odds ratio of twenty-six point one for major complications. The confidence interval was very wide, from three point three to two hundred and five, because the numbers were small, but the signal is clear. In the same study, a focal lesion and treatment with low molecular weight heparin were independent risk factors for bleeding. On the other hand, low-dose aspirin in the five days before the biopsy did not increase bleeding. The odds ratio was zero point nine six.

Young infants are the second group. In 2005, Azzam and colleagues reported that eighteen percent of infants under three months had a complication after biopsy. But seven of the twelve complications were caused by the sedation, not the needle. Sedation events occurred in ten percent of these infants, against two point six percent of older children. So in the youngest infants, the main danger is the sedation. This is why many centres now do the biopsy in young infants under local anaesthesia alone.

Children with cancer or after a bone marrow transplant are the third group. All three deaths in the Cohen series were in this group. And in a series from Oshrine and colleagues, four of five complications after transjugular biopsy occurred in children after a stem cell transplant.

An open biopsy of a tumour is the fourth. In the Children's Oncology Group study of one hundred and twenty-one children with hepatoblastoma, thirty-six percent of children with an open biopsy were transfused, against none of those with a percutaneous biopsy.

The fifth risk is doing the biopsy together with another procedure. In one thousand one hundred and twenty-two biopsies after liver transplant in Japan, published by Sanada and colleagues in 2021, complications occurred in three point one nine percent when the biopsy was combined with another procedure, and in zero point four nine percent when it was done alone.

The sixth is a bile duct problem after transplant. In a study of four hundred and fifty-one biopsies after transplant, by Perito and colleagues in 2019, five of the nine severe events were cholangitis, and they happened in children with a biliary stricture. This is why antibiotics are given before a biopsy when there is a Roux loop or a known stricture.

Finally, some diseases carry their own bleeding tendency. The ESPGHAN paper mentions Alagille syndrome, and the syndrome of arthrogryposis, renal dysfunction and cholestasis, called ARC syndrome, in which the platelets do not work properly.

Other complications are rare. Pneumothorax was reported in only one of four hundred and sixty-nine children. Perforation of the bowel, bile leak, bleeding into the bile ducts and a fistula between an artery and a vein can all occur, but mostly as case reports. A fistula should be suspected if a child develops a new abdominal bruit, or new signs of portal hypertension after a biopsy.

Part seven. When complications show.

How long should we watch a child after a biopsy? This matters, because it decides whether the biopsy can be done as a day case.

In adult series, sixty percent of complications appear within two hours, and ninety-six percent within twenty-four hours. Fatal complications usually happen within six hours.

In children, the best data come from Melbourne. In 2017, Bolia and colleagues looked at six hundred and twenty-six paediatric biopsies. Every complication was recognised within eight hours. They calculated that an eight-hour observation period, instead of an overnight stay, would save about sixty hospital beds a year.

In the Oslo study, a routine ultrasound on the next day found unsuspected bleeding in two point six percent of children, but none of them needed treatment.

So it is the observation window, not the overnight bed, that makes a day-case biopsy safe. The ESPGHAN paper asks for bed rest for at least one hour, observation for at least six hours, and no contact sport for one week. The British guideline asks for at least three hours of observation.

What should you watch for? Pain, a fast heart rate, falling blood pressure, pallor, difficulty breathing, fever, or bleeding from the biopsy site. And remember how pain looks at different ages. An infant cannot tell you that it hurts. An infant who is bleeding may simply be irritable. An older child often complains of pain in the right shoulder, because blood under the diaphragm irritates the phrenic nerve. If pain persists or gets worse, do an urgent ultrasound.

Part eight. The numbers before the needle.

We now come to the most controversial part of this chapter. What should we do with the INR, the platelet count and the fibrinogen before a liver biopsy? Here the guidelines do not only differ. On several points they directly contradict each other.

A short word on units. Platelet counts here are given in units of ten to the ninth per litre, which is the same number as thousands per microlitre.

Let us start with the INR. The ESPGHAN paper says that if the INR is above one point five, it should be corrected with fresh frozen plasma, at twelve to fifteen millilitres per kilogram, or with recombinant factor seven a, or the biopsy should be done by the transjugular or laparoscopic route. The British guideline says that if the INR is above one point four, a non-lesional biopsy should be done by the transvenous route. But the American and European liver societies, in their coagulation guidelines of 2020 and 2022, say that the INR should not be corrected before procedures. These positions contradict each other.

The same is true for plasma. ESPGHAN gives plasma immediately before the biopsy if the INR stays high. The British guideline says not to give plasma to correct an INR of two or less. And the American and European societies say no prophylactic plasma at all.

For platelets, the guidelines differ. ESPGHAN recommends a transfusion immediately before the biopsy if the platelet count is sixty or less. The British guideline suggests the transvenous route if the count is below fifty, and says that a platelet transfusion, or a drug that raises the platelet count, called a thrombopoietin receptor agonist, can be considered. The American society does

not recommend routine correction, and does not recommend these drugs. The European society says that below fifty, it should be decided case by case.

For fibrinogen, the guidelines also differ. ESPGHAN recommends giving fibrinogen if it is below one gram per litre. The British guideline says that a fibrinogen below zero point six grams per litre, together with platelets below thirty or an aPTT above one hundred seconds, means a very high risk of bleeding. The American and European societies do not recommend routine correction.

There is also a contradiction about the risk of the procedure itself. The American coagulation guidance classes a percutaneous liver biopsy as a high-risk procedure for bleeding. The European guidance classes it as low risk.

There are only two points where the documents come close. The first is vitamin K. ESPGHAN recommends parenteral vitamin K at least six hours before the biopsy, and the British guideline makes a strong recommendation to correct vitamin K deficiency. The second is aspirin, and here they only partly agree. ESPGHAN says that low-dose aspirin can be continued after liver transplant. The British guideline suggests stopping it for three days if there is time.

How can the guidelines disagree so much? Because the evidence that the INR predicts bleeding is weak, and it is especially weak in children.

In the Los Angeles study, the children who bled had a median age of one point eight months. The children who did not bleed had a median age of eighty-four months, which is seven years. The clotting tests did not predict bleeding. The best predictors were young age, a low haematocrit and the reason for the biopsy. In the Melbourne study of six hundred and twenty-six biopsies, bleeding was not related to the INR either.

Then there is a randomised trial from Delhi, published by Verma and colleagues in 2026. It included children with decompensated cirrhosis who needed a biopsy while their INR was between two and two and a half. Half of them had their transfusions guided by a bedside clotting test called rotational thromboelastometry, or ROTEM. The other half had standard care, based on the INR and the platelet count. In the ROTEM group, seventy-six percent received a transfusion, against one hundred percent in the standard group. But the volume transfused was not lower, the bleeding was no different, and the ROTEM strategy cost more. So ROTEM reduced the number of children exposed to blood products, but not the bleeding.

And in adults with cirrhosis, prophylactic transfusion before procedures did not protect them. Bleeding occurred in nineteen percent of those who were transfused, against seven percent of those who were not. This does not prove that

transfusion causes bleeding, because sicker patients are more likely to be transfused. But it certainly does not show a benefit.

Why does the INR fail? Because in liver disease, the liver makes less of the proteins that prevent clotting as well as less of those that promote it. The balance is reset at a lower level, but it is still a balance. The INR measures only one side of it. We discussed this rebalanced haemostasis in the chapter on synthetic function.

Perhaps the most honest sentence is in the ESPGHAN paper itself. It says that no data on a safe INR threshold have been published.

So what should you do in practice? Give vitamin K. Know your local protocol, because it will usually follow one of these guidelines. And when the clotting tests are clearly abnormal, the safest answer is often not to correct the numbers, but to change the route, to a plugged or a transjugular biopsy.

Part nine. The paediatric biopsy, step by step.

Let us now walk through a biopsy, following the ESPGHAN protocol.

Before the procedure, do an ultrasound. Look for ascites, dilated bile ducts, a haemangioma, or bowel lying between the liver and the diaphragm. Check a full blood count and an INR within twenty-four hours. Give intravenous vitamin K at least six hours before. The child should fast for about four hours. And if the child is at risk of hypoglycaemia, as many infants with liver disease are, start a glucose infusion during the fast.

For the pass itself, the child lies on the back, with the right arm behind the head. After sedation or anaesthesia, inject a long-acting local anaesthetic, such as bupivacaine, into the skin and down the track. Use real-time ultrasound. Use a sixteen gauge needle if the size of the child allows. And make no more than three passes.

Now comes a step that is easy to forget. The core must be divided at the bedside, once, into the right containers. There is no second chance. Put the main part into formalin for routine histology. Snap-freeze a piece at minus eighty degrees Celsius, for molecular studies, enzymes or metabolic tests. Put a small piece into glutaraldehyde for electron microscopy, if a storage disease or a transporter defect is possible. And if Wilson disease is possible, keep a separate core, not put in formalin, for measuring copper. This is why the question must be clear before the needle goes in.

After the biopsy, press firmly on the site. The child should stay in bed for at least one hour, and be observed for at least six hours. Give antibiotics if there is a Roux loop, a biliary stricture or cholangitis. And advise no contact sport for one week.

For infants under three months, remember that the sedation is the main risk. Many centres use local anaesthesia alone in this age group.

Part ten. What the guidelines say.

How do the guidelines compare on the practical side of biopsy? There are seven points worth knowing.

First, the size of the core. Here the guidelines differ. ESPGHAN asks for twenty millimetres by one point eight millimetres, at least ten portal tracts or at least two centimetres in biliary atresia, and at least six portal tracts in a wedge biopsy. The American standard asks for at least two centimetres with a sixteen gauge needle and at least eleven complete portal tracts. The British guideline asks for at least twenty millimetres with a sixteen gauge needle, and the Royal College of Pathologists sets a minimum of ten millimetres and six portal tracts.

Second, image guidance. Here the guidelines agree. ESPGHAN says that ultrasound reduces complications and is mandatory after a segmental transplant. The American fatty liver guidance of 2025 accepts the percutaneous, EUS-guided or transjugular routes. The British guideline recommends ultrasound where possible, or at least imaging within the last three months.

Third, the route in a child with abnormal clotting. Here the guidelines contradict each other, as we saw in part eight. ESPGHAN corrects an INR above one point five or changes the route. The British guideline goes transvenous above an INR of one point four or below a platelet count of fifty. The American society calls the biopsy high risk but does not correct the INR. The European society calls it low risk.

Fourth, antibiotics. Here the guidelines broadly agree. ESPGHAN recommends them for cholangiopathy after transplant, for heart valve disease and for bacteraemia. The British guideline recommends them for a biliary-enteric anastomosis, a previous infection or biliary sepsis.

Fifth, training. Here the numbers differ. The ESPGHAN panel suggests at least twenty supervised biopsies. The American society asks for at least forty. The North American paediatric society, NASPGHAN, asks for twenty, with half of them in children under three years. The United Kingdom curriculum asks for fifty.

Sixth, acute liver failure. Here the guidelines agree. ESPGHAN calls the role of biopsy limited and questionable, and suggests the transjugular route only if autoimmune hepatitis or Wilson disease is suspected. The British guideline says to avoid biopsy in decompensated disease or kidney injury, and to use the transjugular route if the biopsy is essential.

Seventh, infants. Only the paediatric document addresses them. It warns about the risk of sedation under three months, and notes that local anaesthesia alone is common.

Part eleven. Four habits, and choosing the route.

Let us bring this together into four habits.

The first habit is to count portal tracts, not centimetres. Six portal tracts can make a diagnosis. Eleven are needed to stage fibrosis. A transjugular core has a median of only four. And if the core looks adequate at the bedside but the report says it was fragmented, talk to the laboratory.

The second habit is to remember that infants are different. In the youngest infants, the sedation is the main risk. Globules in alpha one antitrypsin deficiency and paucity in Alagille syndrome appear late. And in infants under ten kilograms, consider plugging the track.

The third habit is to remember that the INR misleads. No safe INR threshold has been published. Young age predicts bleeding better than the INR. And ROTEM reduces the number of children who receive blood, but not the bleeding.

The fourth habit is to let the context decide. In acute liver failure, the biopsy rarely changes care. For a mass, use a percutaneous core, not an open biopsy. And with a Roux loop, give antibiotics.

Put simply, before the needle goes in, you should be able to answer four questions. Which question am I asking? Which route will I use? How many portal tracts do I need? And where will each piece of the core go?

To finish the teaching part of the chapter, here are the common clinical situations, and the route and sample for each one.

For a cholestatic infant who may have biliary atresia, use an ultrasound-guided percutaneous biopsy, and weigh the risk of sedation. Aim for at least ten portal tracts, and repeat the biopsy if it was taken early and is equivocal. Never delay the Kasai operation for it.

For cholestasis with a normal GGT, use a percutaneous biopsy. Ask for a BSEP stain, put a piece into glutaraldehyde for electron microscopy, and freeze a piece. And stage the cirrhosis before any biliary diversion.

For suspected autoimmune hepatitis or autoimmune sclerosing cholangitis, use a percutaneous biopsy, or a transjugular one if the child is coagulopathic. Aim for eleven portal tracts, ask for a CK seven stain, and do a magnetic resonance cholangiogram as well. Remember to biopsy again before stopping treatment.

For possible Wilson disease, use a percutaneous or transjugular biopsy, and take a separate unfixed core for copper. Remember that cholestasis also raises the liver copper.

For fatty liver disease with an uncertain diagnosis, use an ultrasound-guided or EUS-guided biopsy, aim for more than two centimetres, and ask the pathologist to report the zonal pattern. Obesity did not lower the yield in the blind biopsy series.

For acute liver failure or decompensated cirrhosis, use the transjugular route, and only if the result will change treatment. Take at least two cores, and measure the portal pressure at the same sitting. Remember the odds ratio of twenty-six for major complications.

For a graft that is not working after transplant, use an image-guided percutaneous biopsy, or a transjugular one for a small graft with ascites. Rejection needs fewer portal tracts to diagnose than fibrosis needs to stage. And with a Roux loop or a stricture, give antibiotics.

For a liver mass, use a percutaneous core passed through normal liver, and sample both the lesion and the background liver. Remember that thirty-six percent of children with an open biopsy were transfused.

Part twelve. Four clinical cases.

Case one. A five-week-old boy has jaundice and pale stools. His conjugated bilirubin and GGT are high. The ultrasound shows a small gallbladder, but the triangular cord sign is not clear. The team plans a liver biopsy under general anaesthesia next week, because the anaesthetic list is full. What do you advise? This child may have biliary atresia, and time matters. The biopsy is useful, but it must not delay the Kasai operation. At five weeks, the typical features may not yet be fully developed, so an equivocal result should lead to a cholangiogram, not to waiting. Many centres would do this biopsy under local anaesthesia with light sedation, or combine it with the cholangiogram in theatre. Ask for a core with at least ten portal tracts. Before the biopsy, give intravenous vitamin K and check the glucose during the fast.

Case two. An eleven-year-old girl has an ALT of six hundred units per litre, a high IgG and a positive antinuclear antibody. Her INR is one point seven, and it does not improve after vitamin K. The team wants to confirm autoimmune hepatitis. How should the biopsy be done? A biopsy is recommended for every child with suspected autoimmune hepatitis. The INR after vitamin K means that this is liver synthetic failure, not vitamin K deficiency. Under ESPGHAN, an INR above one point five means either correcting it or changing the route. Under the British guideline, an INR above one point four means the transvenous route. The

American and European societies would not correct the INR. A reasonable choice is the transjugular route, taking at least two cores and knowing that they may contain fewer portal tracts. Wilson disease must also be excluded in this age group, so keep a separate unfixed core for copper.

Case three. A four-year-old boy had a left lateral segment liver transplant, with a Roux loop, two years ago. His GGT is rising. The team wants a biopsy to look for rejection. What precautions are needed? After a segmental graft, the normal landmarks do not apply, so the biopsy must be image-guided. Because he has a Roux loop, he needs antibiotics, and a rising GGT should raise the question of a biliary stricture, which was the setting for cholangitis in the Perito series. Do an ultrasound first, and consider imaging of the bile ducts. Do not combine the biopsy with another procedure unless it is necessary, because this raised the complication rate about six-fold in the Sanada series. Rejection can be diagnosed on fewer portal tracts than fibrosis needs for staging.

Case four. A thirteen-year-old boy with chronic hepatitis B has a raised ALT. His biopsy report says, "mild portal inflammation, fibrosis stage one". The core is one centimetre long and contains five portal tracts. The team concludes that the liver disease is mild. Do you agree? Not yet. Five portal tracts may be enough to describe the inflammation, but they are not enough to stage fibrosis. We need at least eleven. As we saw in the last chapter, a short core systematically makes the disease look milder than it is. The stage of one should be read as "at least stage one". Before deciding against treatment, combine the biopsy with non-invasive tests such as elastography, and consider a repeat biopsy with a sixteen gauge needle if the decision depends on the stage.

Key points to remember.

Before every biopsy, ask what the tissue will tell you that genes, enzymes and imaging cannot.

Biliary atresia remains the main diagnostic indication, but the biopsy must never delay the Kasai operation. Alpha one antitrypsin deficiency does not need a biopsy for diagnosis.

Today the biopsy more often answers how severe a disease is and whether treatment works. Every child with suspected autoimmune hepatitis should have one.

In Wilson disease, the key tissue test is quantitative copper on a separate unfixed core. A liver copper above two hundred and fifty micrograms per gram of dry weight is diagnostic.

Ultrasound guidance is the default, and it is mandatory after a segmental graft. With abnormal clotting, change the route to a plugged or transjugular biopsy.

Transjugular cores are thinner, with a median of four portal tracts. Take at least two cores.

Six portal tracts can make a diagnosis. Eleven are needed to stage fibrosis. Most paediatric cores are only about one centimetre long.

About one in five first biopsies is non-diagnostic. Short cores, vague indications and fragmentation in the laboratory are the main causes. A sixteen gauge needle gives more tissue without more bleeding.

Sampling error is predictable. It is common in young infants, in patchy disease such as acute liver failure, and in tissue from under the capsule.

Major bleeding occurs in about one to three percent of children. The risk belongs to the child: infants, children with cancer, and children with liver failure.

In infants under three months, the sedation causes more complications than the needle.

All paediatric complications in the Melbourne series appeared within eight hours. Observe for at least six hours. An irritable infant or an older child with right shoulder pain may be bleeding.

The guidelines contradict each other on correcting the INR. The INR does not predict bleeding in children, and no safe threshold has been published. Give vitamin K.

Divide the core at the bedside, once: formalin, frozen, glutaraldehyde, and an unfixed core for copper.

Questions to test yourself.

Question one. Which disease still needs a biopsy for diagnosis in most centres, and what is the one rule about it? Answer: biliary atresia. The biopsy must never delay the Kasai operation, and the cholangiogram is the final test.

Question two. Is a liver biopsy needed to diagnose alpha one antitrypsin deficiency in a cholestatic infant? Answer: no. The diagnosis is made by the phenotype or by testing the SERPINA one gene. In early infancy, the typical globules are often not yet visible on the biopsy anyway.

Question three. How many portal tracts are needed to make a diagnosis, and how many to stage fibrosis? Answer: about six to make a diagnosis, and at least eleven to stage fibrosis.

Question four. What is the main advantage and the main disadvantage of the transjugular route? Answer: it is safer when clotting is abnormal, because bleeding goes back into the circulation. But the cores are thinner and contain fewer portal tracts, a median of four in the Lucknow study, so at least two cores should be taken.

Question five. In infants under three months, what causes most complications of a liver biopsy? Answer: the sedation. In the Azzam series, seven of twelve complications came from sedation. Many centres use local anaesthesia alone in this age group.

Question six. How long should a child be observed after a percutaneous biopsy, and why? Answer: at least six hours, according to ESPGHAN. Most complications appear within a few hours, and in the Melbourne series all paediatric complications were recognised within eight hours.

Question seven. A child's INR is one point six before a biopsy. What do the guidelines say? Answer: they disagree. ESPGHAN would correct the INR or change the route. The British guideline would use the transvenous route above one point four. The American and European societies would not correct the INR. And no study has shown a safe INR threshold in children.

Question eight. Wilson disease is possible in a child having a biopsy. What must be done at the bedside? Answer: take a separate core, not put into formalin, for quantitative copper measurement. A value above two hundred and fifty micrograms per gram of dry weight is diagnostic.

End of chapter.